

EXPERIMENT -04

Aim

To implement and demonstrate the working of the ID3 (Iterative Dichotomiser 3) Decision Tree algorithm using a dataset stored in a CSV file and classify a new sample using the generated decision tree.

Procedure

1. Open Google Colab and upload the dataset CSV file.
2. Import the required Python libraries such as Pandas, LabelEncoder, and Decision Tree Classifier.
3. Read the dataset from the CSV file.
4. Convert categorical attributes into numerical values using Label Encoding.
5. Separate the input features and target class.
6. Build the decision tree model using the ID3 algorithm by using Entropy as the splitting criterion.
7. Train the model using the given dataset.
8. Visualize the generated decision tree.
9. Give a new sample input to the trained model.
10. Predict and display the class of the new sample.

Program

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier, plot_tree
import matplotlib.pyplot as plt

from google.colab import files

uploaded = files.upload()

Choose Files PlayTennis.csv
PlayTennis.csv(text/csv) - 432 bytes, last modified: 7/16/2026 - 100% done
Saving PlayTennis.csv to PlayTennis.csv

import pandas as pd

df = pd.read_csv("PlayTennis.csv")

df
```

Output:

	Outlook	Temperature	Humidity	Wind	PlayTennis
0	Sunny	Hot	High	Weak	No
1	Sunny	Hot	High	Strong	No
2	Overcast	Hot	High	Weak	Yes
3	Rain	Mild	High	Weak	Yes
4	Rain	Cool	Normal	Weak	Yes
5	Rain	Cool	Normal	Strong	No
6	Overcast	Cool	Normal	Strong	Yes
7	Sunny	Mild	High	Weak	No
8	Sunny	Cool	Normal	Weak	Yes
9	Rain	Mild	Normal	Weak	Yes
10	Sunny	Mild	Normal	Strong	Yes
11	Overcast	Mild	High	Strong	Yes
12	Overcast	Hot	Normal	Weak	Yes
13	Rain	Mild	High	Strong	No

Result:

Thus, the ID3 Decision Tree algorithm was successfully implemented using the PlayTennis CSV dataset in Google Colab. The decision tree was generated using entropy-based information gain, and the trained model successfully classified a new sample.